

**MANUAL
FOR APPLYING
WALLTOPIA PRELIMINARY LOADS
ON EXISTING STRUCTURES**

PRELIMINARY LOADS – NOT FOR CONSTRUCTION

I. Climber loads

Climber loads according to: EN12572-1: Artificial climbing structures. Safety requirements and test methods for ACS with protection points

All load values are given with characteristic values.

Safety factors according to acting national standard.

II. Worked example for usage of Walltopia table for preliminary loads

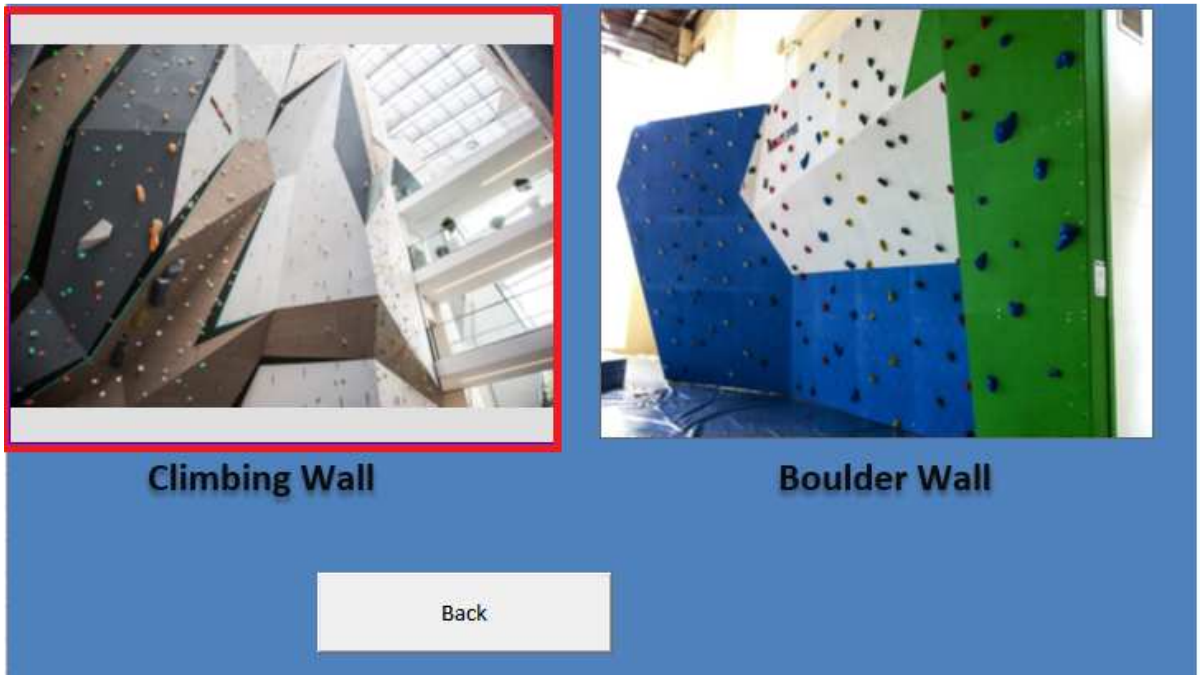
II.1. Example 1

1. Choose preferred units:

EU (kN; m; kNm)
USA (Lb; ft; lbft)

For this example we will use the metric system.

2. Next choose the type of climbing structure:



The loads imposed by climbers differ between the two variants so this option must be chosen adequately.

For the two options to be understood:

Climbing Wall = Artificial climbing wall **with** protection points

Boulder Wall = Artificial climbing wall **without** protection points

From herein all climbing structures will be referred to as ACS (Artificial climbing structure)

For this next step we will choose the option for artificial climbing wall **with** protection points.

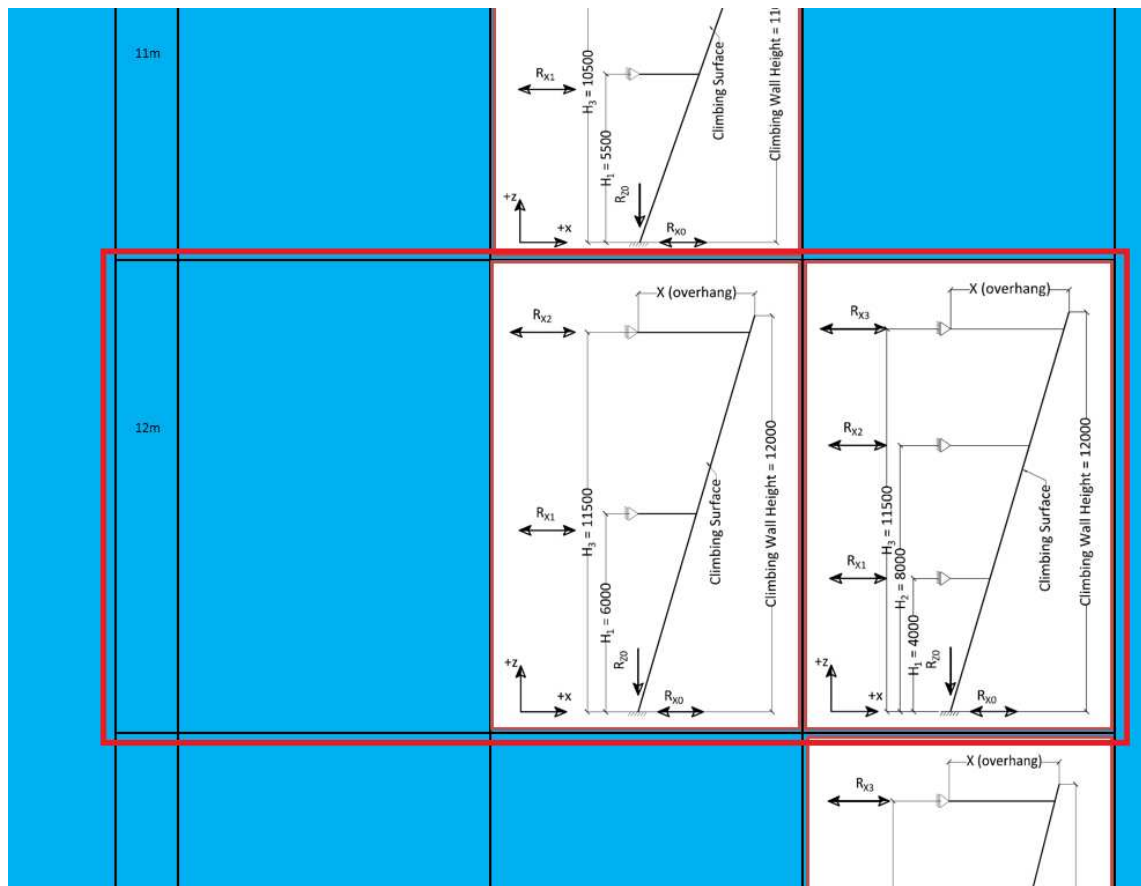
3. Next page is a table containing a variety of schemes for ACS

	one attachment	two attachments	three attachments
8m			
9m			
10m			
11m			
12m			

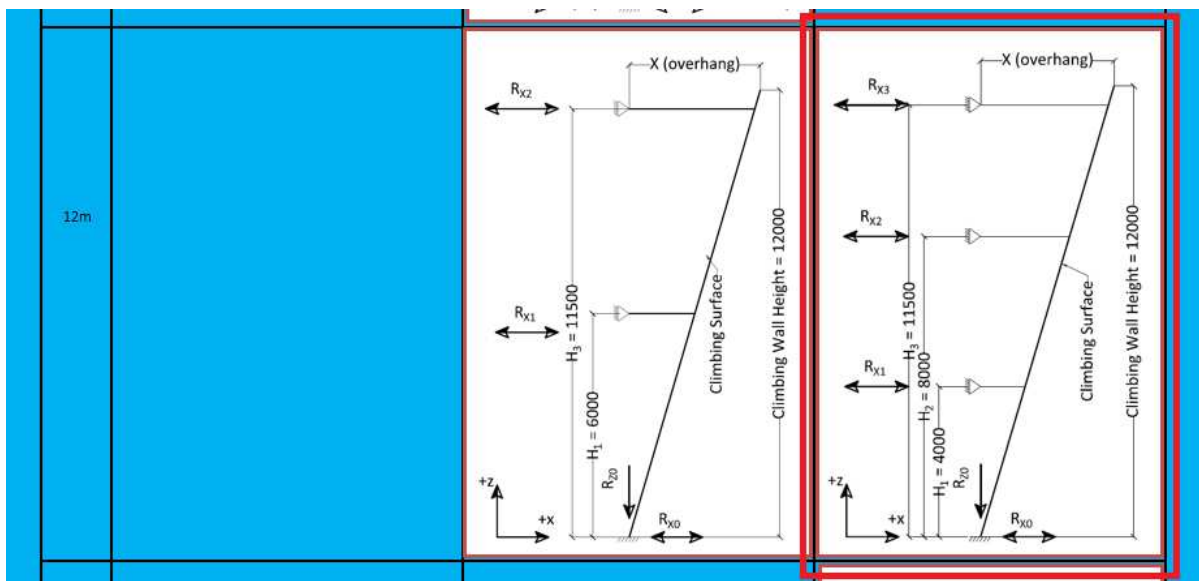
Back

Horizontal values correspond to the **number of attachment points** by height of the climbing wall. Vertical values correspond to the height of the climbing surface.

For this example we will choose an ACS with height of the climbing surface of 12m.

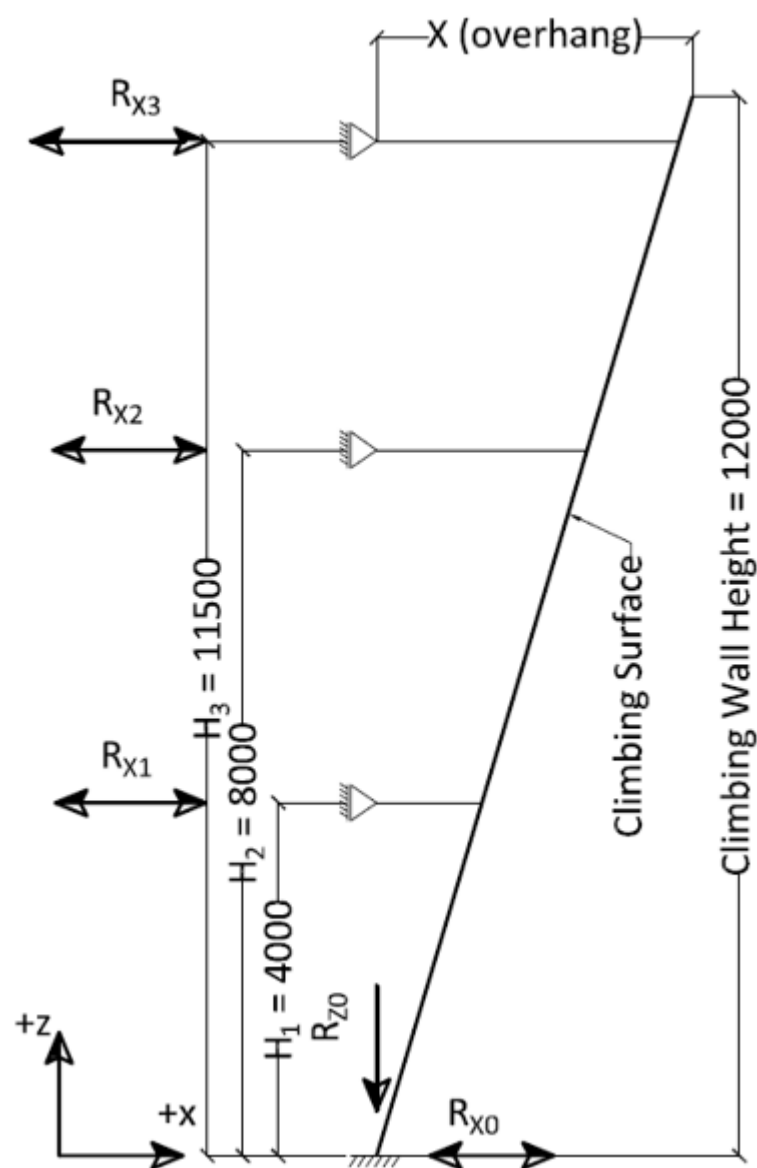


There are two possible schemes of attachment for this height – attachment on two levels and attachment on three levels. For the example choose attachment on **three levels**.



On the next page we have two schemes to illustrate how the loads are imposed on the existing structure.

This first scheme shows the positive directions of the local coordinate system and the force loads designation for **a single section of the ACS**.

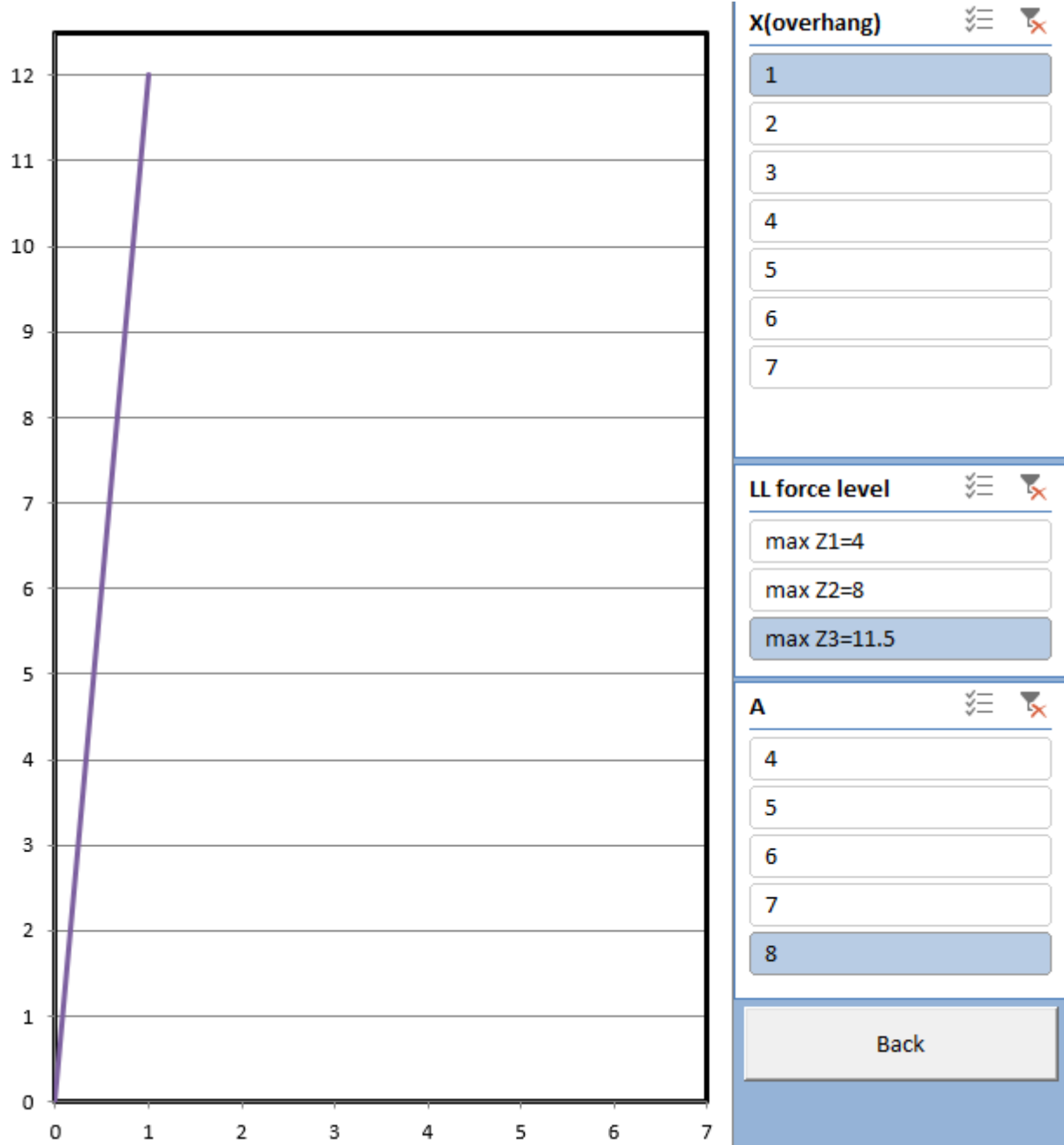


The load values corresponding to one section of the climbing wall are given in the table located in the upper left.

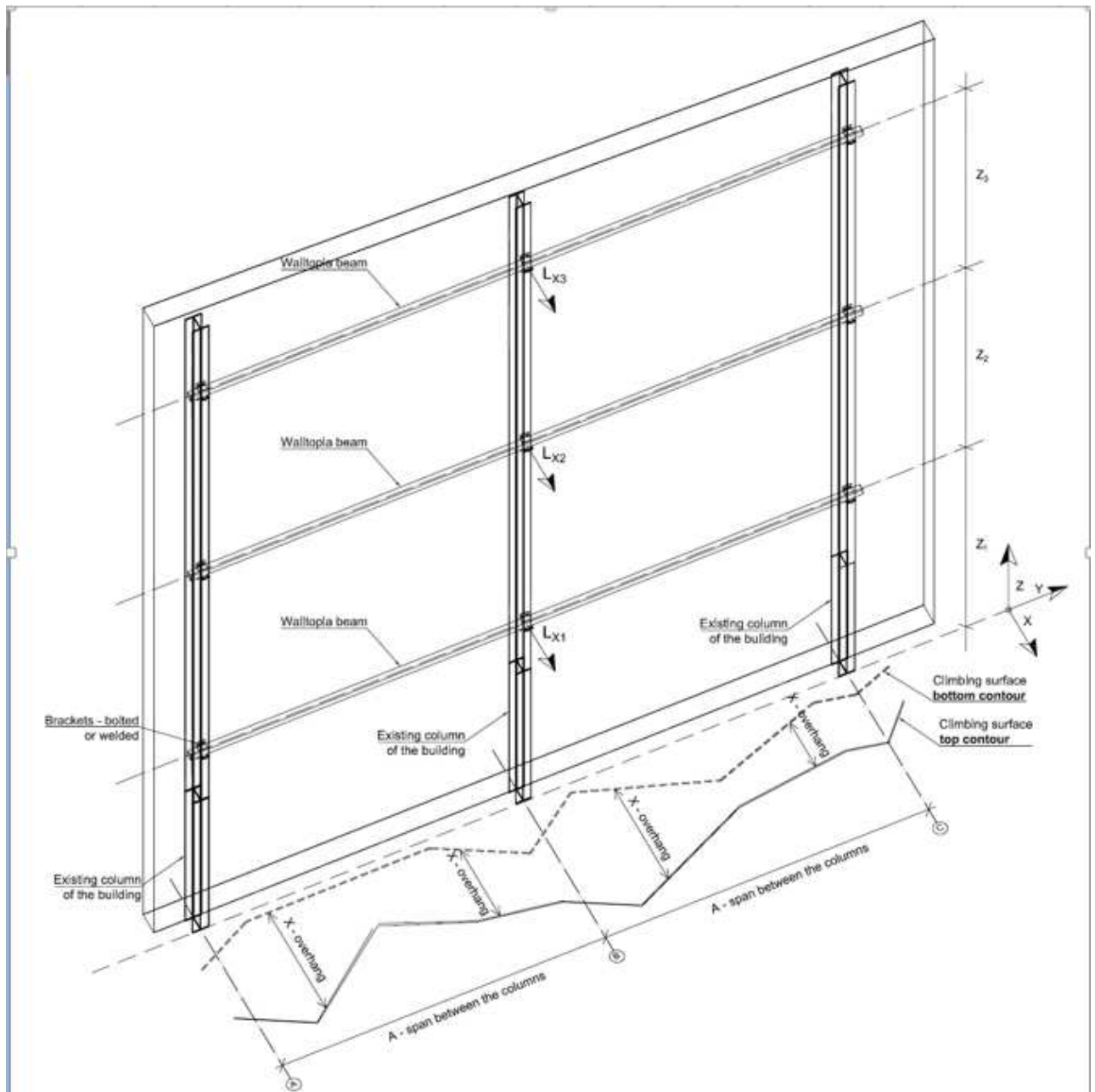
RZ0DL	RX0DL	RX1DL	RX2DL	RX3DL	RZ0LL	RX0LL	RX1LL	RX2LL	RX3LL	
-4,4	-0,7	0,5	0,3	-0,1	0,3	-6,8	-0,9	1,2	-2,8	3,8

Where all vales under R represent load values for one axis of the ACS (applicable for base points and single axis attachment points), while all values under L represent the load values on points of attachment to building columns usually spanned by a beam by Walltopia.

On the left of the first picture we have a graph representing the overhang value of the ACS, on the right we have the input buttons to regard.



On the right the second picture illustrates how forces **from the climbing structure** are imposed on the existing building.



Values L_{xi} shown in pictures are the imposed loads on columns of the building, with columns span length taken into consideration

X(overhang)



1
2
3
4
5
6
7

X – represents the length of the overhang of the ACS. There is a correlation between the overhang value and the horizontal components of the load values. The greater the overhang, the greater the load values. The overhang is the difference in length between the bottom contour and the top contour of the climbing wall.

LL force level



max Z1=4
max Z2=8
max Z3=11.5

Force level represents the elevation at which we want to obtain the maximal values of the load imposed on the existing structure. Since live load point of application is variable, we need to define maximal values for each point of attachment in elevation

NOTE: For any given point the maximal load values shall be taken for a single level of attachment at a time (according to standard two or more climbers can not fall down simultaneously), on other level(s) of attachment the loads shall be taken with their corresponding values for each situation under consideration.

A

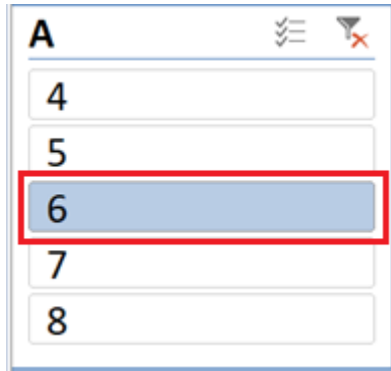


4
5
6
7
8

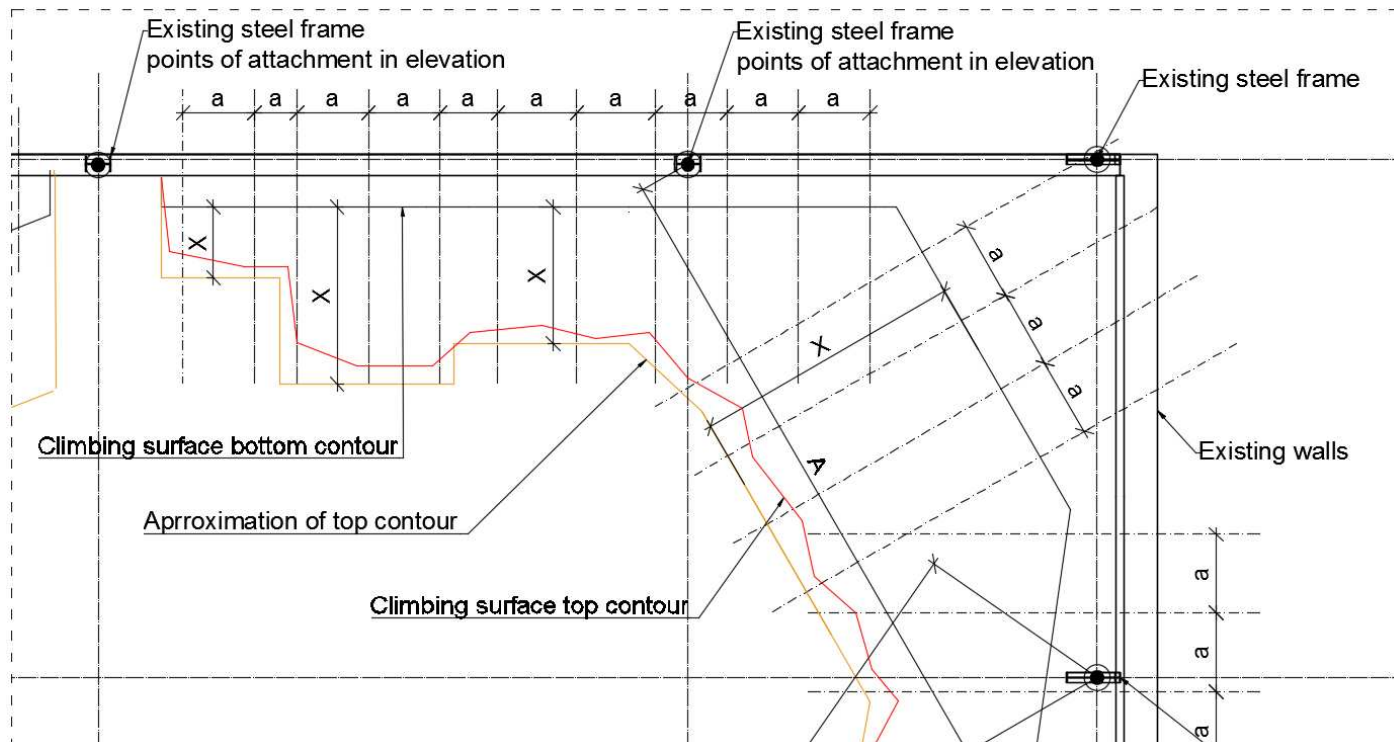
A – represents the span between the columns of the building. The span length affects the load values for the dead load cases. The greater the span, the higher the horizontal loads from dead load cases.

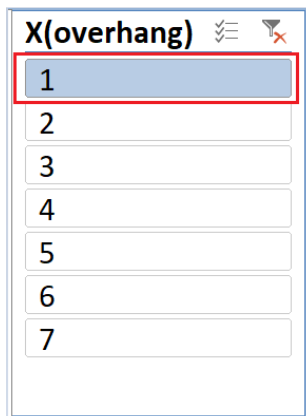
III. How to correctly define input data and apply the loads to building

1. After having chosen the button in correspondence the specific ACS height, we have to choose the span length of our building columns as shown:



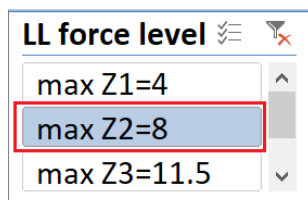
2. Having chosen span length of the columns of the building, we then have to approximate a value of the overhang of the ACS. We have to take in consideration areas of the ACS in plan with a smooth transition of the overhang and approximate a value as shown in the subsequent picture.





X(overhang)
1
2
3
4
5
6
7

3. After making an approximation of all possible overhang lengths, we have to select for the specific column (frame) we want to design and choose the respective overhang
4. As a result of the variable points at which the live load may be applied the force level button is added . In terms of live load we must consider multiple points of application of the live load, and in term consider multiple maximum values for a single column (frame). In the following example we want to view the local maximum for a 12m ACS with an overhang of 1m, in a building with a column raster of 6m.



LL force level
max Z1=4
max Z2=8
max Z3=11.5

We have selected the necessary elevation at which we would like to extract the maximum value.

5. Result overview: Dead load

The following table represents all results for the specific case

Important: all values given in this table are example and are not to be used for design. For real results please refer to table.

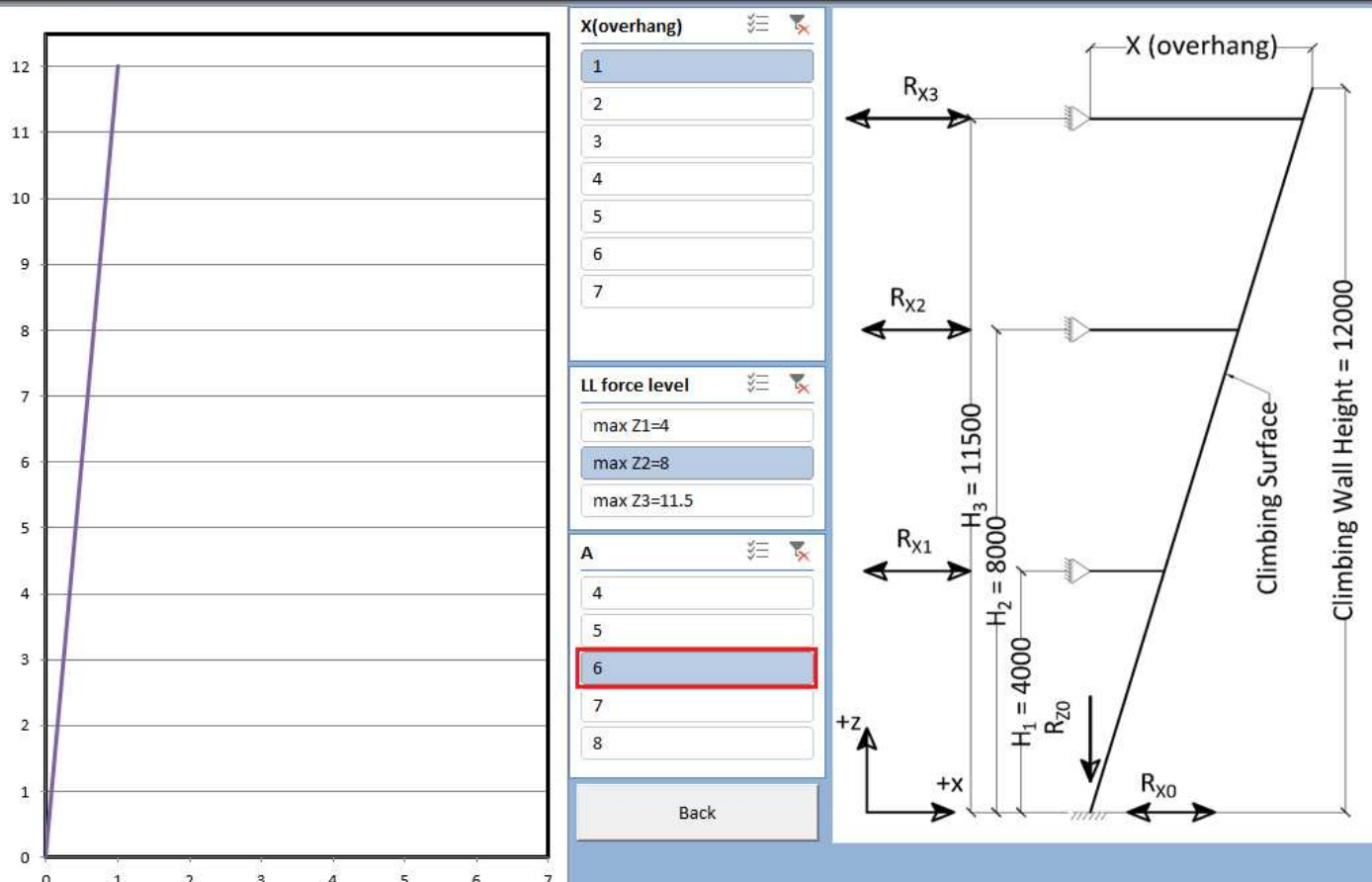
RZ0DL	RX0DL	RX1DL	RX2DL	RX3DL	RZ0LL	RX0LL	RX1LL	RX2LL	RX3LL	LX1DL	LX2DL	LX3DL	LX1LL	LX2LL	LX3LL
-4.4	-0.7	0.5	0.3 / -0.1	0.3	-6.8	-0.7	-0.8	2.2	0.6	3	1.8 / -0.6	1.8	-2	5.5	1.5

The highlighted values represent the results from dead load on existing structure.

R values show the load value for 1 axis of the ACS

L values show the load value for the columns of the build so we will focus on them

RZ0DL	RX0DL	RX1DL	RX2DL	RX3DL	RZ0LL	RX0LL	RX1LL	RX2LL	RX3LL	LX1DL	LX2DL	LX3DL	LX1LL	LX2LL	LX3LL		
-4,4	-0,7	0,5	0,3	-0,1	0,3	-6,8	-0,7	-0,8	2,2	0,6	3	1,8	-0,6	1,8	-2	5,5	1,5
Grand Total																	



At the all elevations the highlighted loads must be applied to the building columns (frames).

Dead load values are a function of column span.

All load values in tables are with their characteristic values and **MUST** be factored according to the current acting standard

DEAD LOAD acts upon the whole building factored as per the local acting standard

6. Result overview: Live load

The following table represents all results for the specific case

Important: all values given in this table are example and are not to be used for design. For real results please refer to table.

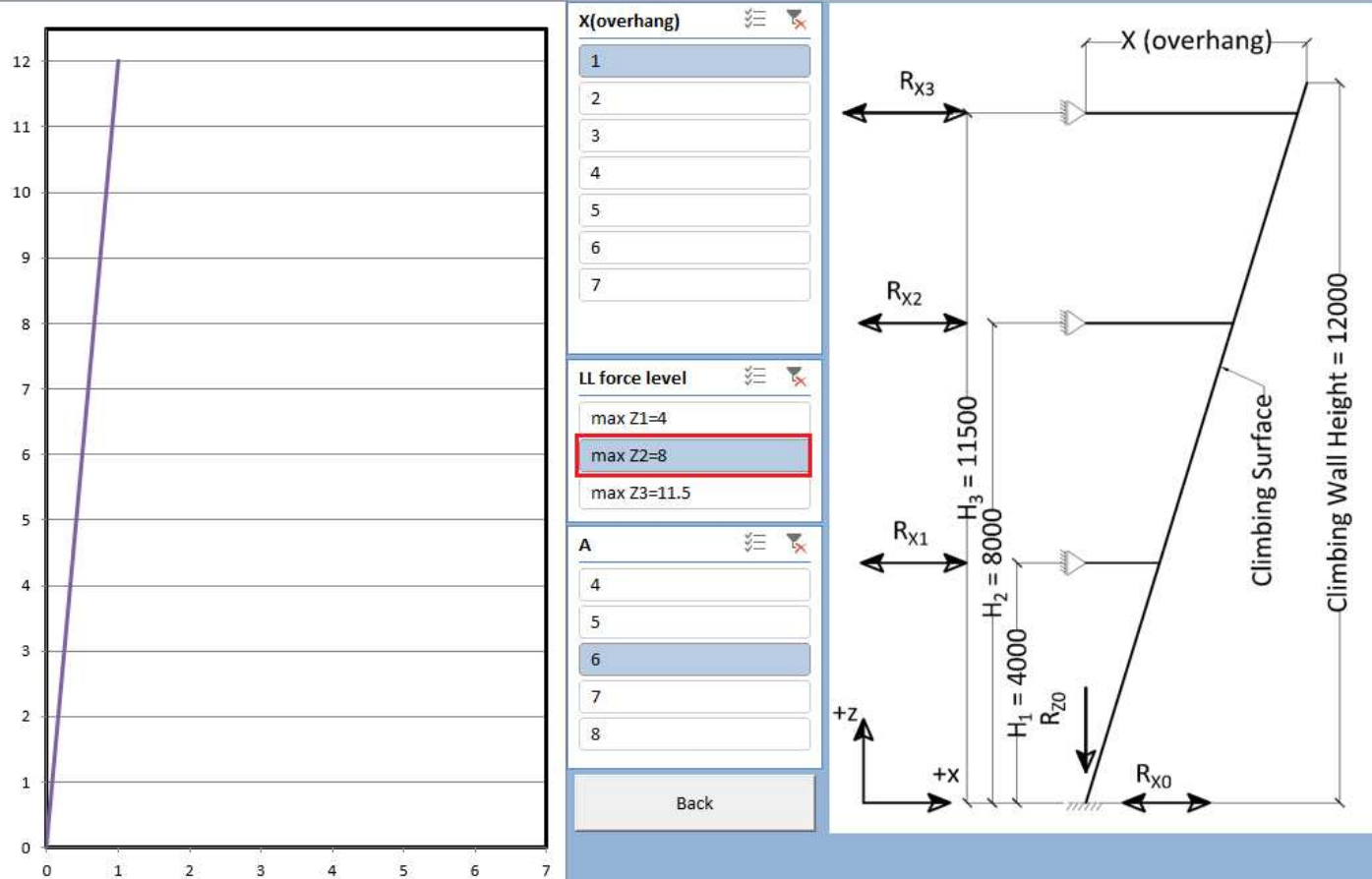
RZ0DL	RX0DL	RX1DL	RX2DL	RX3DL	RZ0LL	RX0LL	RX1LL	RX2LL	RX3LL	LX1DL	LX2DL	LX3DL	LX1LL	LX2LL	LX3LL
-4,4	-0,7	0,5	0,3 / -0,1	0,3	-6,8	-0,7	-0,8	2,2	0,6	3	1,8 / -0,6	1,8	-2	5,5	1,5

The highlighted values represent the results from live load on existing structure.

R values show the load value for 1 axis of the ACS

L values show the load value for the columns of the build so we will focus on them

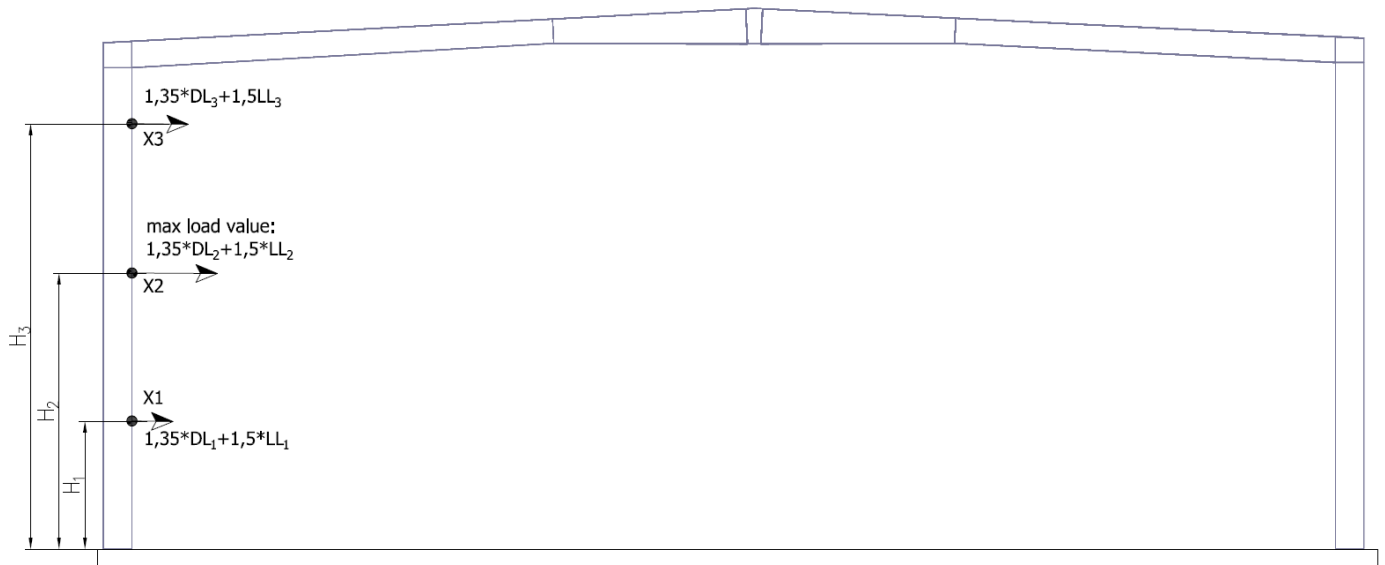
RZ0DL	RX0DL	RX1DL	RX2DL	RX3DL	RZ0LL	RX0LL	RX1LL	RX2LL	RX3LL	LX1DL	LX2DL	LX3DL	LX1LL	LX2LL	LX3LL
-4,4	-0,7	0,5	0,3 / -0,1	0,3	-6,8	-0,7	-0,8	2,2	0,6	3	1,8 / -0,6	1,8	-2	5,5	1,5
Grand Total															



At the necessary elevation the highlighted loads must be applied to the building columns (frames).

This is the maximal value for the point of attachment X2, along with its concurrent values. All load values in tables are with their characteristic values and MUST be factored according to the current acting standard

In the following drawing we see the application of the loads on a 2D steel frame

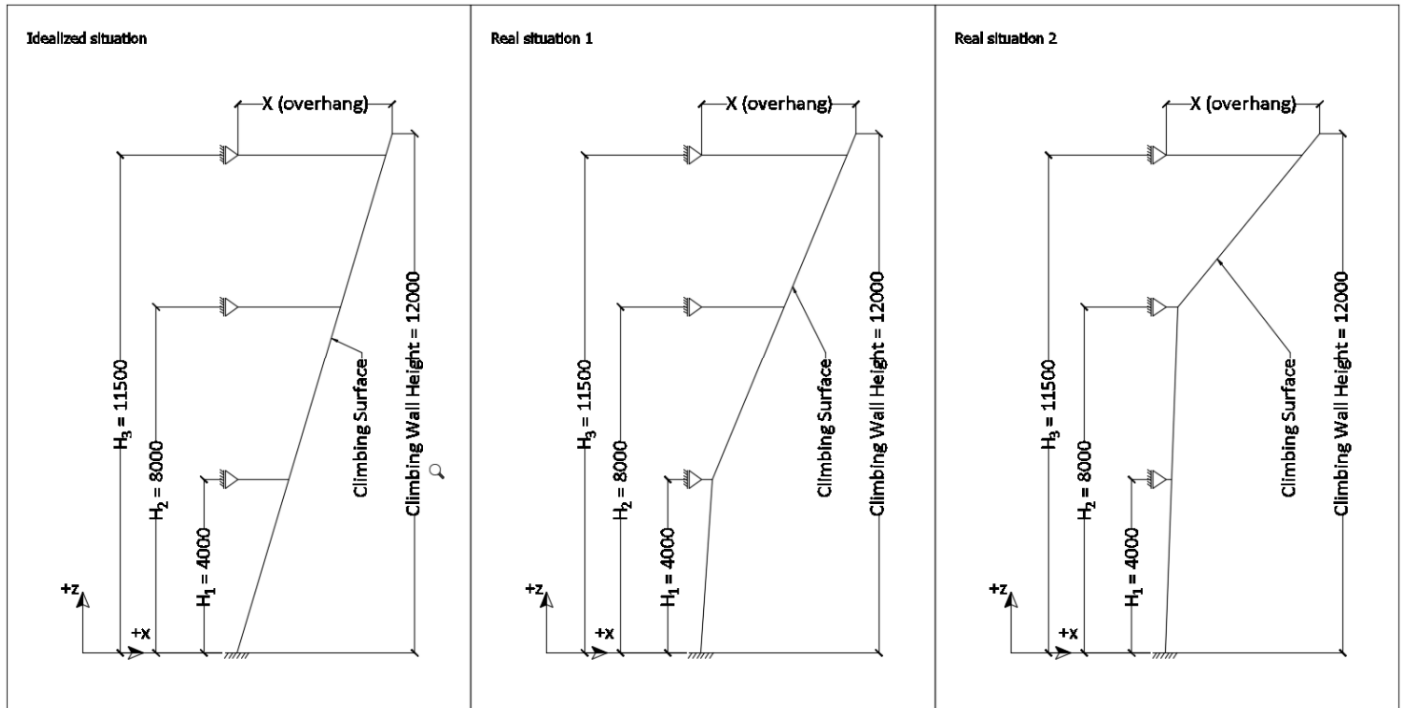


The following procedure must be repeated for each point of attachment as shown previously, in order to determine the prevail design values for building frame.

Important Note:

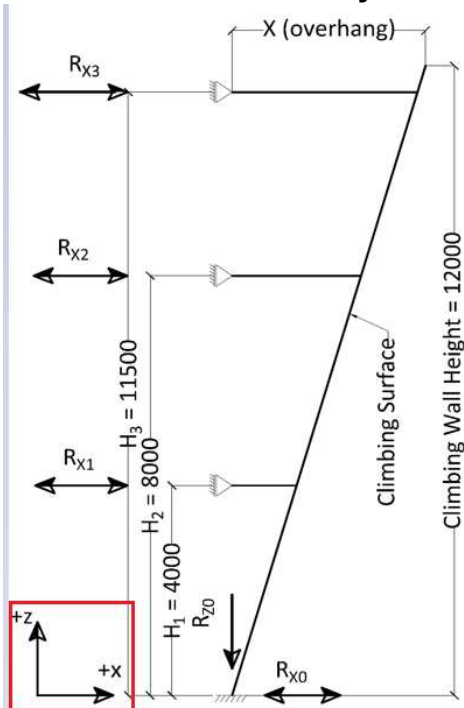
LIVE LOAD does NOT act simultaneously on each building frame, as where maximum value is applied, load in neighbouring frames have negligible values. Each frame must be regarded as individual. In cases of end columns biaxial loads may occur. Please refer to the special notes for additional explanation.

Note: If a load is ambiguous (has both positive and negative values), it is up the designer to choose which situation is most unfavourable for his case. The reason for ambiguity in some values is the difference in climbing surfaces geometry.



As we can see, the three climbing surfaces have the same overhang, but different geometry. The tables take into consideration a multitude of cases, which is why some concurrent values have both positive and negative signs.

The values in tables are loads on the building NOT reactions, and their positive direction is that of the coordinate system



IV. Special cases

The scheme below represents an example arrangement of a ACS which are attached to existing structure. It aims to provide a clarification on how to apply the preliminary loads in special cases, such as:

1. Existing Columns / Frames at the corners of the building (specified as “corner columns” in the document below)
2. Existing Columns / Frames at the end of the climbing wall / boulder structure (specified as “end columns” in the document below)

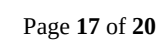
Key Notes:

A – span between existing steel columns /frames.

a – span between the columns of the climbing wall/ boulder structure. The average distance to be taken between 0.9 – 1.2m, however it may varies depending on the design and / or existing building arrangement.

X – overhang of the climbing wall/boulder. The most unfavourable or average overhang in close proximity to the attachment point should be used when calculating the loads for identified existing column / frame.

L_x – characteristic load applied on the existing column / frame at identified attachment level (varies, i.e. x_1 , x_2 , x_3). The level of the attachment point is as given in Walltopia design table.



IV.1. Loads imposed on corner columns

Reference: Preview 1, as shown on the scheme above

Notes:

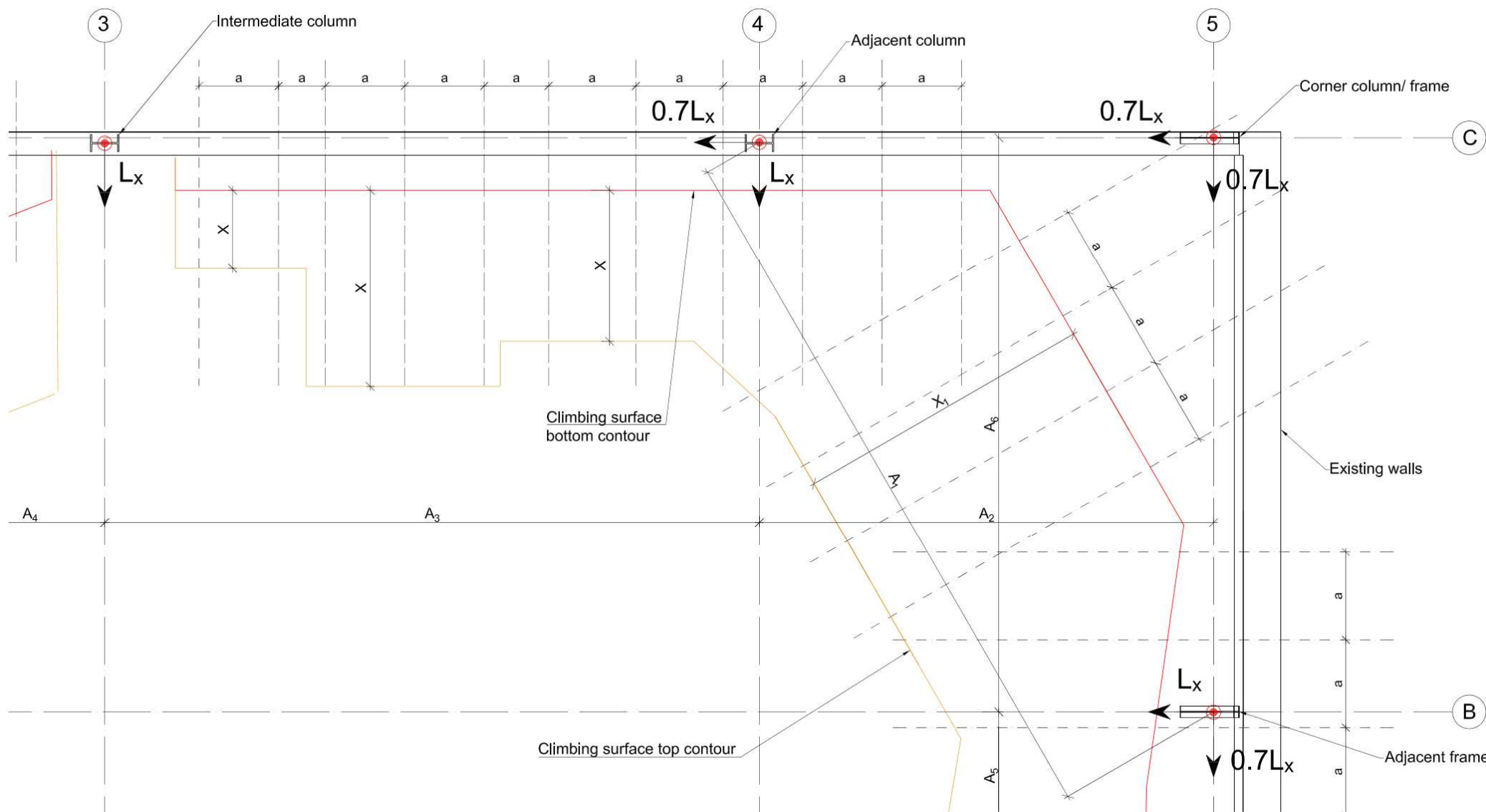
For the existing columns / frames located in the corners of the building, the load (L_x) should be split into two components so that their effect is the same along both global axes. The load to be multiplied by a coefficient ($k = 0.7$) in both directions.

The load (L_x) on both adjacent columns is also to be split and multiplied by coefficient 0.7 however, only in the direction perpendicular to the column's axis as shown on the scheme below. Preview 1. This is done to consider the effect of load distribution at the corners onto the adjacent columns.

When calculating the load L_x for the columns at the corners the span $A = A_1$ between the two adjacent columns, as shown on the scheme below.

When calculating the load L_x for the adjacent columns the span $A = 0.5 \cdot (A_2 + A_3)$, resp. $A = 0.5 \cdot (A_5 + A_6)$, an average value of the distances between the axis to the intermediate column and the column at the corner.

When calculating the load L_x for the intermediate columns the span $A = 0.5 \cdot (A_3 + A_4)$, an average value of the distances between the axis to the neighbouring columns / frames.



IV.2. Loads imposed on end columns

Reference: Preview 2, as shown on the scheme above

Notes:

For existing columns / frames located at the end (or at closest proximity to the end) of the climbing wall / boulder structure, the load (L_x) is to be multiplied by a coefficient $k = 0.5$. This is due to the fact that the loads imposed at the end columns act only within one half length of the span ($=0.5 \cdot A$) to the adjacent column. The other half of the load is imposed on the adjacent column within the limits of the climbing wall/boulder.

Preview 2

